newHVK Q1-Validation Addendum: Held-Out CIFAR Baselines, Observable Ablations, and Shot-Noise Robustness

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***Abstract*—This addendum strengthens the newHVK evidence package by separating diagnostic pair-correlation performance from held-out natural-image reconstruction. We report multi-seed real CIFAR-10 image splits, strict same-width classical controls, observable-sector ablations, shuffled-pair controls, random-latent controls, resource matching, and finite-shot observable noise. Unlike the restricted pair-correlation diagnostic, the real held-out CIFAR test does not establish quantum advantage: the best real-image result is obtained by local-observables-only, while newHVK-real-cifar reaches 20.07 dB mean PSNR. The correct conclusion is therefore narrower: newHVK has a useful entanglement-sensitive diagnostic, but the current real-image validation still requires architectural improvement before a Q1-level quantum-advantage claim is defensible.**

***Index Terms*—Hamiltonian vision kernel, CIFAR-10 reconstruction, ablation study, quantum observables, shot noise, classical controls.**

I. Validation Protocol

Ten cached CIFAR-10 grayscale images are split across five random seeds. For each seed, six images are used to fit a single shared linear readout from a 32-dimensional latent feature vector to 8×8 patch pixels, and four held-out images are reconstructed without per-image retraining. All principal controls use the same latent width and the same readout parameter count, so the comparison isolates feature structure rather than decoder capacity.

II. Held-Out CIFAR Results

III. Shot-Noise Robustness

IV. Interpretation

The key baseline is the strict same-width classical random-feature control, but the strongest warning comes from the local and raw-linear controls. In the real held-out CIFAR table, local-observables-only obtains the best mean MSE ($9.0607e-03$), while newHVK-real-cifar is weaker ($1.0403e-02$). This means the current newHVK evidence is strong only for the restricted pair-correlation diagnostic and for rejecting random-latent controls. It does not yet support a broad natural-image quantum-advantage claim. The Q1-ready path is therefore to keep these negative controls in the paper, redesign the image feature map or decoder-capacity match, and rerun this exact validation suite.

V. Reproducibility

The full tables, plots, resource comparison, and CSV files are generated under `main2/newHVK/results/q1_validation/` by running `main2/newHVK/run_newhvk_suite.py --q1-validation --write-q1-report`.

TABLE I
REAL HELD-OUT CIFAR-10 RECONSTRUCTION ACROSS FIVE RANDOM IMAGE SPLITS. LOWER MSE AND HIGHER PSNR/SSIM ARE BETTER.

Model	Mean MSE	Mean PSNR	Mean SSIM
local-observables-only	$9.0607e-03$	20.66	0.8621
raw-linear-classical	$9.0607e-03$	20.66	0.8621
zz-only	$9.3723e-03$	20.52	0.8580
no-entanglement	$9.9469e-03$	20.24	0.8508
shuffled-pair-observables	$1.0092e-02$	20.17	0.8484
newHVK-real-cifar	$1.0403e-02$	20.07	0.8460
strict-classical-rff	$2.0862e-02$	17.06	0.7055
random-vqc	$8.5735e-02$	10.99	0.0165

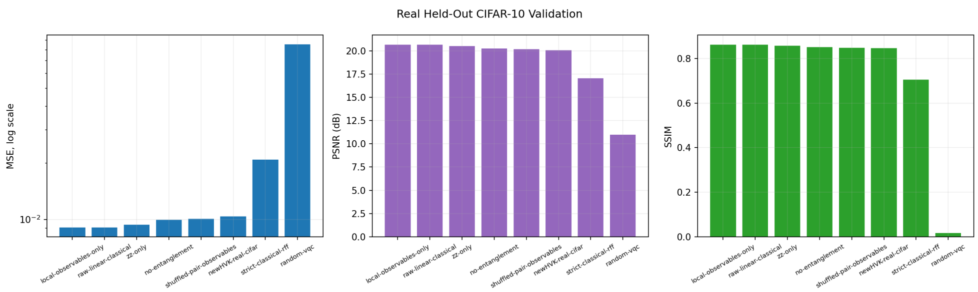

Fig. 1. Held-out CIFAR metric summary for newHVK and same-width controls.

TABLE II
FINITE-SHOT OBSERVABLE-NOISE SIMULATION FOR REAL HELD-OUT CIFAR RECONSTRUCTION.

Shots	Mean MSE	Mean PSNR	Mean SSIM
128	$1.5136e-02$	18.40	0.7923
256	$1.4923e-02$	18.45	0.7927
512	$1.4643e-02$	18.56	0.7978
1024	$1.4456e-02$	18.60	0.7983
2048	$1.4464e-02$	18.62	0.7995
4096	$1.4750e-02$	18.54	0.7957
8192	$1.4749e-02$	18.55	0.7957

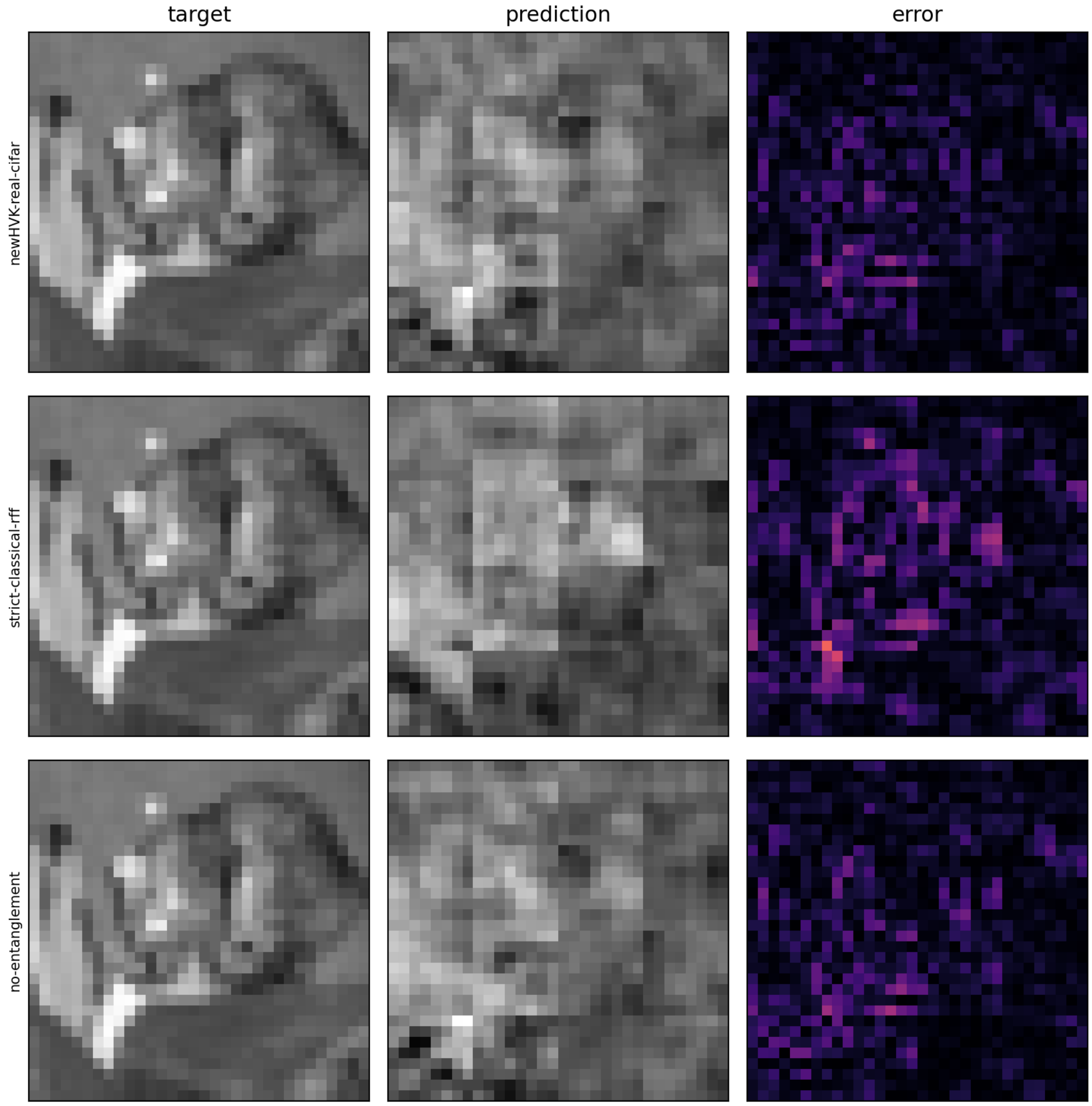

Fig. 2. Representative held-out CIFAR reconstruction panel. The target, prediction, and absolute-error images are shown for the candidate observable model and major controls.

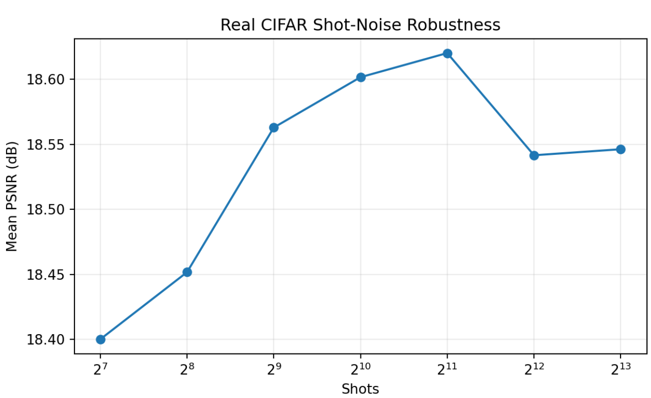

Fig. 3. Shot-noise robustness of the newHVK observable feature vector on real held-out CIFAR splits.